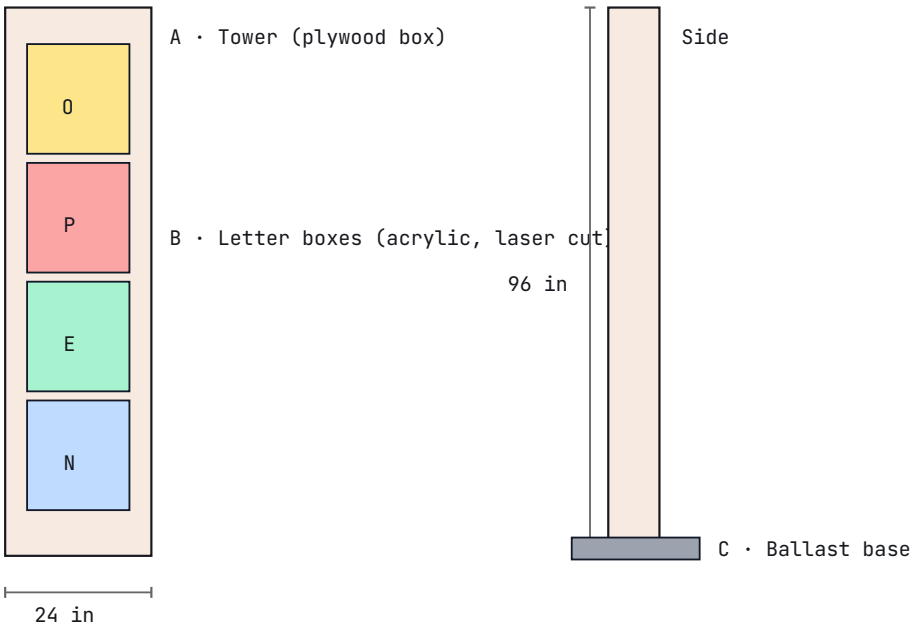
BUILD SPEC · NOVEMBER 2026 · SAMPLE, FICTIONAL PROJECT

# Night Market sign

An 8 ft freestanding sign that spells OPEN LATE in color-changing letters, for an outdoor night market, running all night on one battery.

| 2 × 8 × 0.7 ft | $1,205 | 294 W | 4 days |
|---|---|---|---|
| OVERALL SIZE | MATERIALS | POWER | BUILD TIME |
| | estimate incl. tax, Nov 2026 | 10.4 h on a 3.6 kWh battery | 2 people |

## 01 Overview




| | |
|---|---|
| A · Tower | 1/2 in birch plywood box on a 2×2 frame, track-saw cut, glued and screwed |
| B · Letters | Eight laser-cut acrylic boxes (white face, black returns) with LED strip inside |
| C · Base | Plywood plinth with four sandbags and ratchet straps to a stake or fence |

## 02 Bill of materials

| PART | QTY | UNIT | EACH | TOTAL | SOURCE | NOTES |
|---|---|---|---|---|---|---|
| **Structure** | | | | | | |
| 1/2 in birch plywood 4×8 | 3 | sheet | $68.50 | $205.50 | link | cabinet grade |
| 2×2 poplar 8 ft | 6 | piece | $9.80 | $58.80 | link | inner frame |

| PART | QTY | UNIT | EACH | TOTAL | SOURCE | NOTES |
|---|---|---|---|---|---|---|
| **Face** | | | | | | |
| 1/4 in white acrylic 24×18 | 7 | sheet | $34.00 | $238.00 | local plastics supplier | letter diffusers |
| 1/8 in black acrylic 24×18 | 7 | sheet | $22.00 | $154.00 | local plastics supplier | letter returns |
| **Hardware** | | | | | | |
| #8 × 1-1/4 in wood screws (100) | 1 | box | $9.99 | $9.99 | | |
| Ratchet straps and sandbags | 4 | set | $18.00 | $72.00 | | wind ballast |
| **Electrical** | | | | | | |
| WS2815 LED strip 5 m 12 V | 4 | reel | $38.00 | $152.00 | | per-pixel color |
| Pixel controller (Wi-Fi) | 1 | each | $60.00 | $60.00 | | |
| 12 V 30 A power supply (weatherproof) | 1 | each | $45.00 | $45.00 | | |
| Inline fuse holder + 30 A fuses | 2 | each | $7.50 | $15.00 | | one per supply output |
| 14 AWG two-conductor wire 50 ft | 1 | spool | $32.00 | $32.00 | | |
| **Finish** | | | | | | |
| Exterior primer and black satin paint | 1 | kit | $58.00 | $58.00 | | two coats |
| **Materials subtotal** | | | | **$1,100.29** | | |
| Tax (9.5%) | | | | $104.53 | | |
| **Total** | | | | **$1,204.82** | | |

## 03 Cut list and sheets

| PART | QTY | SIZE (IN) | MATERIAL | PROCESS |
|---|---|---|---|---|
| Side panel | 2 | 8 × 96 | 1/2 in ply | Track saw |
| Back panel | 2 | 24 × 96 | 1/2 in ply | Track saw |
| Letter face | 8 | 16 × 20 | 1/4 in white acrylic | Laser |
| Letter return | 8 | 16 × 20 | 1/8 in black acrylic | Laser |

### Sheet goods

| MATERIAL | SHEET SIZE (IN) | PART AREA (SQ FT) | SHEETS (+15% WASTE) |
|---|---|---|---|
| 1/2 ply | 48 × 96 | 44.0 | 2 |

| MATERIAL | SHEET SIZE (IN) | PART AREA (SQ FT) | SHEETS (+15% WASTE) |
|---|---|---|---|
| 1/4 white acrylic | 24 × 18 | 17.8 | 7 |
| 1/8 black acrylic | 24 × 18 | 17.8 | 7 |

Sheet counts come from part area plus a waste allowance; real nesting can need one more sheet. Check the layout in your nesting or laser software.

## 04 Files and machine settings

| PROCESS | FILES | MATERIAL | STARTING SETTINGS | MACHINE TIME |
|---|---|---|---|---|
| Laser | letters.svg: red = cut, blue = score | 1/4 in cast acrylic | Cut 100% power, 8 mm/s, 2 passes TEST ON SCRAP | 3.5 h |
| Laser | returns.svg | 1/8 in cast acrylic | Cut 90% power, 14 mm/s TEST ON SCRAP | 2 h |

## 05 Power

### Power budget

| DEVICE | QTY | WATTS EACH | WATTS |
|---|---|---|---|
| WS2815 LED strip 5 m 12 V | 4 | 72 | 288 |
| Pixel controller (Wi-Fi) | 1 | 6 | 6 |
| Total draw | | | 294 W |
| With 25% headroom | | | **368 W** · 30.6 A at 12 V |
| Runtime on 3,600 Wh battery (85% usable, at total draw) | | | 10.4 h |

12 V DC throughout. 30 A supply fused at the output, 14 AWG runs under 15 ft, controller in the base behind a removable panel.

## 06 Build steps

1. **Tower:** cut panels, glue and screw the 2×2 frame.
   Check: square within 1/16 in before the back goes on.
2. **Letters:** laser the faces and returns, bond with acrylic cement.
   Check: dry-fit every letter in its cutout.
3. **LEDs:** lay strip inside each letter box, solder joins, heat-shrink.
   Check: test every letter before sealing.
4. **Finish:** prime and paint the tower, two coats.
   Check: 24 h cure before install.

## 07 Safety

**Before you build**

- Cut acrylic only: never laser PVC or vinyl. Run the laser with ventilation and stay with it.
- An 8 ft sign catches wind: ballast with at least 100 lb and strap it to a fixed point; take it down above 25 mph.
- Fuse the battery output and keep all wiring 12 V DC.

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